1. Why is os.replace on a temp file in the same folder atomic, while writing directly to summary.json is not?

os.replace is safer because it replaces the old file with the new file in one step. If the program stops while creating the temp file, the old summary is still safe. But if we write directly to summary.json, the file might become incomplete if the program suddenly stops.

2. Why does the manifest use a content hash instead of the file name or modification time?

The content hash checks the actual content of the file. Even if the file name or modification time changes, the hash will stay the same if the content is unchanged. This helps the system detect duplicate files more accurately.

3. Why is Decimal required for money? Show a float calculation from this lab that gives a wrong total.

Decimal is required because it gives more accurate results when adding money. Float can sometimes give a small difference because of how it stores decimal numbers. The correct total should be 0.30, so using Decimal is better for money calculations.

4. Explain how your generator keeps memory constant on the 1,000,000-row file. What would the peak be if you used list(csv.DictReader(f))?

The generator creates and writes one row at a time instead of keeping all rows in memory. This means it can process 1,000,000 rows without using a lot of memory.

If I used list(csv.DictReader(f)), all 1,000,000 rows would be stored in memory at once. This would use much more RAM and could make the program slow or cause it to run out of memory.

5. Why quarantine an unreadable file instead of deleting it or stopping the whole run?

Quarantining is better because it keeps the unreadable file so we can check it later. It also allows the system to continue processing the other files. Deleting it would lose the data, while stopping the whole run would prevent other files from being processed.